

Cameco Corporation

World's Largest Uranium Producer — Nuclear Fuel Supply Chain Chokepoint

PhD Due Diligence Report | BLRI-DD-CCJ-AUG2026

13 August 2026 | Blue Lotus Intelligence | CivilisationOS

Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Cameco: The Saudi Aramco of Uranium — Owning the Fuel for the Nuclear Renaissance

Cameco Corporation is the world's largest publicly traded uranium producer, operating the Cigar Lake (Saskatchewan) and McArthur River/Key Lake mining complexes — among the highest-grade uranium deposits on Earth. Cigar Lake ore grades of 15%+ U₃O₈ are 10-15× the world average, making Cameco's operations the lowest-cost and highest-quality uranium production on the planet. In Seldon ξ -variable terms: uranium is the fuel for the nuclear renaissance. As AI datacentres drive demand for 24/7 firm carbon-free power, nuclear is the only large-scale answer — and Cameco owns the mines.

Cameco acquired a 49% stake in Westinghouse Electric (nuclear reactor technology + services) through a consortium with Brookfield Renewable, transforming Cameco from a pure mining company into an integrated nuclear fuel cycle company: uranium mining → fuel fabrication → reactor technology. This vertical integration mirrors the Seldon model of chokepoint control across the full value chain.

Source: moomoo OpenD [2026-08-13] · Cameco 40-F SEC filing · World Nuclear Association uranium market data

PART II — QUALITATIVE ANALYSIS

Cigar Lake: The World's Highest-Grade Operating Uranium Mine

Cigar Lake (Cameco 54.5% operator) produces uranium at ore grades 15-18% U3O8 — the highest-grade operating mine in the world. The ore body requires robotic jet boring in frozen ground (freeze curtain technology) to handle the radiation levels. No competitor can replicate Cigar Lake's grade; it is a geological accident that Cameco controls. Production capacity of 18 million pounds/year (100% basis) makes it the world's largest high-grade uranium mine.

Westinghouse Acquisition: Vertical Integration into Reactor Technology

The Cameco/Brookfield consortium acquired Westinghouse Electric for \$7.875B in 2023. Westinghouse manufactures nuclear fuel assemblies, operates reactor services, and is developing the AP300 SMR. Cameco's 49% stake in Westinghouse gives it exposure to reactor services revenue that is recurring (refuelling, maintenance contracts on 50% of global nuclear capacity) — a fundamentally different revenue profile from cyclical uranium mining.

Source: Cameco 40-F EDGAR · Westinghouse acquisition materials · WNA uranium market report

PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]
Price: \$99.070 | Prev Close: \$99.030 | Mkt Cap: \$43.148B
EPS TTM: \$0.968 | Net Income: \$421.6M | Div Yield: 0.17%
52w: \$68.821–\$135.240 | PE TTM: 170.52x | Shares: 435.5M

Metric	Value	Source
Price (2026-08-13)	\$99.070	moomoo OpenD
Market Cap	\$43.148B	moomoo OpenD
EPS TTM	\$0.968	moomoo OpenD
Net Income TTM	\$421.6M	moomoo OpenD
52-Week Range	\$68.821–\$135.240	moomoo OpenD
Div Yield	0.17%	moomoo OpenD

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch

PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Mkt Cap	vs CCJ
Kazatomprom (KAP)	Largest uranium producer (Kazakhstan)	~\$8B US-listed	CCJ Western-aligned; KAP geopolitical risk (Russian supply chain)
Energy Fuels (UUUU)	US domestic uranium mining	~\$1B	Much smaller; US domestic advantage but not Cigar Lake grade
Uranium Energy Corp (UEC)	US uranium mining + warehousing	~\$2B	US focused; CCJ superior grade + Westinghouse integration
NexGen Energy (NXE)	Arrow deposit development	~\$6B	Next-generation Athabasca deposit; decades from

			production
--	--	--	------------

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The US Utility Nuclear Fuel Buyer (COMPOSITE)

A nuclear fuel procurement manager at a US utility with 8 reactors reviews long-term uranium supply contracts. 'After the Russian invasion of Ukraine, we terminated our contracts with Russian-origin uranium. Cameco's Cigar Lake is the only large-scale Western uranium source that can fill that gap at scale. We signed a 10-year contract with Cameco at \$75/lb U3O8.' Represents the Western utility decoupling from Russian/Kazakh uranium supply that is structurally bullish for Cameco.

COMPOSITE / ILLUSTRATIVE — drawn from US utility nuclear fuel procurement patterns

FINDING F-5-B: Portrait 2: The Kazatomprom-Exposed Uranium Investor Pivoting to CCJ (COMPOSITE)

A uranium fund manager explains the CCJ allocation: 'Kazatomprom is the largest uranium producer globally, but its export route through Russia (Trans-Siberian railway) is sanctioned. Japanese and European utilities cannot take delivery. CCJ is the only liquidity that matters for Western utilities — and they're running at 90%+ capacity at Cigar Lake.' Represents the structural supply shift from Russian/Kazakh to Western uranium.

COMPOSITE / ILLUSTRATIVE — drawn from uranium fund market patterns

FINDING F-5-C: Portrait 3: The Nuclear Renaissance Investor at Uranium Conference (COMPOSITE)

An institutional investor at the World Nuclear Symposium London notes: 'AI datacentre power demand is the demand shock that changes uranium's secular supply/demand balance. Every hyperscaler that signs a nuclear PPA requires more uranium in the fuel cycle. Cameco at 170x PE reflects option value on uranium price going from \$90/lb to \$150/lb — that's where the real upside is.' Represents the bull case: AI-driven nuclear demand → uranium price spike → Cameco earnings leverage.

COMPOSITE / ILLUSTRATIVE — drawn from uranium investor community patterns

PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: Kazakhstan/Russia uranium dominates market at lower cost

EVIDENCE FOR:

Kazatomprom (40%+ global supply) operates at \$20-25/lb production cost vs Cameco's \$35-40/lb; if geopolitical normalisation allows Kazakh uranium back into Western markets, Cameco faces cost disadvantage

EVIDENCE AGAINST:

Western utilities have structural preference for non-Russian supply chain post-Ukraine; US Prohibiting Russian Uranium Imports Act (2024) bans Russian uranium; geopolitical risk is permanent in 5-year horizon

QUANTITATIVE TEST

P(Kazakh/Russian uranium fully competitive in Western markets before 2028)=10%

VERDICT: BEAR — supply reintegration would pressure uranium spot price

Implication: Monitor US/EU sanctions policy on Russian uranium; Kazatomprom Western market access

HH2: HH2: Uranium spot price reaches \$150/lb — Cameco earnings triple

EVIDENCE FOR:

Structural uranium deficit: ~90,000 tonnes/year demand vs ~60,000 tonnes/year primary supply; deficit met by inventory drawdown; as inventories deplete, spot price must rise to incentivise new mines; \$150/lb is the incentive price for new mines

EVIDENCE AGAINST:

Cameco has 70% of its production in long-term contracts at fixed prices (\$55-75/lb); spot price upside is diluted by contracted volumes; benefit is on uncontracted tonnes and contract renegotiations

QUANTITATIVE TEST

P(Uranium spot price >\$150/lb sustained for 6+ months by 2027)=20%

VERDICT: BULL — spot price spike would re-rate CCJ dramatically

Implication: Monitor UxC weekly uranium spot price; utility contracting activity; production guidance

HH3: HH3: Westinghouse acquisition transforms Cameco into nuclear fuel cycle leader

EVIDENCE FOR:

Westinghouse's 50 million+ pounds of annual nuclear fuel assembly fabrication + reactor services = \$2B+ recurring annual revenue for Cameco's 49%; AP300 SMR development adds long-duration growth

EVIDENCE AGAINST:

Westinghouse integration is complex; debt from acquisition; AP300 SMR is years from commercial deployment; Cameco's core competency is mining, not reactor engineering

QUANTITATIVE TEST

P(Westinghouse contributing >\$400M to CCJ net income by FY2027)=35%

VERDICT: BULL — Westinghouse transforms CCJ from pure mining to nuclear infrastructure company

Implication: Monitor Westinghouse quarterly earnings in CCJ disclosures; AP300 SMR licensing status

HH4: HH4: McArthur River/Key Lake production ramp delayed

EVIDENCE FOR:

McArthur River (Cameco 69.8%) was care-and-maintenance from 2018-2022; ramping back up; geological challenges or labour issues could delay production to contracted volumes

EVIDENCE AGAINST:

McArthur River has been restarted and is ramping; Cameco's operational track record is excellent; Cigar Lake provides production diversification

QUANTITATIVE TEST

P(McArthur River production below plan by >15% in any 12-month period)=15%

VERDICT: BEAR — production shortfall forces spot market purchases at premium

Implication: Monitor quarterly production guidance; McArthur River/Key Lake commissioning milestones

HH5: HH5: AI datacentre operators lock in decade-long uranium supply contracts**EVIDENCE FOR:**

Hyperscalers signing nuclear PPAs (Microsoft Three Mile Island, Amazon Susquehanna, Google Kairos) need long-term uranium supply; they may contract directly with Cameco to lock in fuel security

EVIDENCE AGAINST:

Hyperscalers contract with utilities for power, not with uranium miners directly; uranium contracting is utility-level, not end-user level

QUANTITATIVE TEST

P(CCJ direct contract with tech hyperscaler for uranium supply by 2027)=10%

VERDICT: BULL — hyperscaler uranium contracting would be entirely novel demand category

Implication: Monitor any announcements of hyperscaler uranium supply chain involvement

PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: Uranium spot price sustains \$100-120/lb (P = 45%%)

Structural deficit continues; Western utility contracting accelerates; CCJ uncontracted tonnage captures premium; EPS grows to \$2+ by 2027; stock \$110+

Driver: UxC spot price; utility contracting activity; production data

Falsifier: Kazakh supply normalisation; spot price falls to \$60; CCJ de-rates

SCENARIO 2: Westinghouse earnings contribution materialises (P = 35%%)

Westinghouse generates \$400M+ net for CCJ's 49% share; CCJ trades on blended mining + reactor services multiple; de-rates PE to 80-100x on higher earnings

Driver: CCJ quarterly Westinghouse earnings disclosure; Westinghouse revenue growth

Falsifier: Westinghouse integration costs exceed synergies; margin disappointment

SCENARIO 3: Base case: 10% EPS growth — uranium compounder at 170x (P = 40%%)

Cigar Lake + McArthur River running at capacity; long-term contracts deliver stable revenue; Westinghouse growing; EPS \$1.50+ by 2027; stock \$90-120 range

Driver: Annual production guidance; uranium spot price; Westinghouse progress

Falsifier: Spot price collapse + Westinghouse integration issues = 30% selloff

SCENARIO 4: China SMR programme drives uranium demand spike (P = 20%%)

China's 150 reactor construction plan accelerates; Chinese utilities contract globally; uranium demand grows 20%+ above baseline; spot price spike

Driver: China NEA nuclear capacity additions; Chinese utility uranium tendering

Falsifier: China develops domestic uranium mining; reduces global market dependence

SCENARIO 5: Stock retests \$68 low on uranium price collapse (P = 20%%)

Kazatomprom export route normalised; spot uranium falls to \$50/lb; CCJ's long-term contracts protect near-term, but stock de-rates on forward expectations

Driver: UxC spot price; Kazatomprom logistics announcements

Falsifier: New AI nuclear PPA demand spike; utilities rush to contract uranium; price spikes

SCENARIO 6: US Energy Fuels/Uranium royalties compete for domestic contracts (P = 15%%)

US government mandates domestic uranium sourcing for DOE weapons programme; US-only content requirements limit Cameco's US utility market share

Driver: US Prohibiting Russian Uranium Act implementation; US utility uranium tendering

Falsifier: Canada-US uranium trade treated as domestic under CUSMA; CCJ fully qualifies

SCENARIO 7: AP300 SMR licensing with US NRC starts (P = 12%%)

Westinghouse files AP300 SMR NRC design certification; licensing process begins; SMR optionality adds 5-10% to CCJ valuation

Driver: Westinghouse AP300 NRC filing; AP300 design milestone

Falsifier: AP300 licensing delayed past 2028; SMR optionality not valued by market

SCENARIO 8: Japanese utility restart drives Asian uranium demand (P = 25%%)

Japan restarts all 33 pre-approved reactors; Japanese utilities re-enter uranium market for first time since Fukushima; Cameco captures Japan contracts

Driver: Japanese NRA reactor approval schedule; Tepco/KEPCO fuel procurement

Falsifier: Japanese public opposition delays restarts; demand lower than expected

SCENARIO 9: Cameco dividend growth signals confidence (P = 15%%)

CCJ raises dividend from \$0.17% yield to \$0.50%+ yield; signals management confidence in uranium price; institutional income investors add position

Driver: CCJ annual dividend announcement; management guidance

Falsifier: Dividend held flat; cash flow directed to Westinghouse integration debt repayment

SCENARIO 10: Bull case: \$200+ on uranium spike + Westinghouse + AI nuclear (P = 8%%)

Uranium spot \$150+ + Westinghouse \$500M contribution + hyperscaler nuclear contracts; EPS \$3+; CCJ re-rates to 60x earnings = \$180+

Driver: All three catalysts firing; AI nuclear demand spike; uranium supply deficit

Falsifier: Uranium supply normalisation + Westinghouse disappointment + nuclear timeline extension

PART VIII — SELDON VARIABLE DEEP DIVE

Variable	CCJ Impact	Direction	Magnitude
K	Uranium fuel powers nuclear which powers K-acceleration	↑ High	K++ (nuclear fuel supply chokepoint)
ξ	Western uranium supply reduces energy supply concentration	↑ Moderate	ξ+ (supply diversity = equity)
Ω	Western nuclear supply chain reduces Russian/Kazakh energy leverage	↓ Reduces	Ω-- (supply chain geopolitics)
I	Canadian democratic institution with US treaty access	↑ High	I++ (CUSMA ally)

PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
Uranium spot price decline (Kazakh supply normalisation)	LOW (10%)	HIGH	Western sanctions policy protects; long-term contracts buffer
PE at 170x extremely valuation-sensitive	HIGH (inherent)	HIGH	Westinghouse earnings will de-rate PE on higher income
McArthur River production shortfall	MEDIUM (15%)	MEDIUM	Cigar Lake diversifies production; spot market purchases
Westinghouse integration cost overruns	MEDIUM (20%)	MEDIUM	Brookfield operational expertise; Westinghouse has existing revenue
Geopolitical sanctions change — Russia	LOW (10%)	HIGH	US legislative ban (2024) provides legal anchor for 10 years

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: BUY — URANIUM CHOKEPOINT OWNER IN THE NUCLEAR RENAISSANCE

Cameco at \$99.070 (mkt cap \$43.148B) owns the irreplaceable mining assets for the nuclear renaissance — Cigar Lake world's highest-grade ore — plus 49% of Westinghouse's reactor services empire. 170x PE is high but reflects uranium supply chain chokepoint value and earnings leverage to spot price. Western utility decoupling from Russian/Kazakh uranium is structural and permanent post-Ukraine. Seldon alignment: K++ (nuclear fuel = knowledge power foundation), Ω-- (uranium supply geopolitics), I++ (Western democratic energy sovereignty). HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch.